Global Renewable Energy in 2025

How Far Have We Come?

A data-driven overview of renewable energy's growing share in the global energy and electricity mix, spanning hydropower, wind, and solar technologies.

Source: Our World in Data – Renewable Energy | Ember (2026); Energy Institute – Statistical Review of World Energy (2025)

75% of Global GHG Emissions Still Come from Fossil Fuels

75%

of global GHG emissions
come from fossil fuels

5M+

premature deaths per year
from fossil fuel air pollution

- Since the Industrial Revolution, most countries' energy systems have been dominated by fossil fuels.
- Burning fossil fuels for energy is the primary driver of climate change and local air pollution.
- To reduce CO_2 emissions and air pollution, the world must rapidly shift toward low-carbon sources — nuclear and renewable technologies.
- Renewables include solar, wind, hydropower, geothermal, bioenergy, wave, and tidal sources.

⚡

Fossil Fuels

Dominate the
Global Energy System

Since the Industrial Revolution

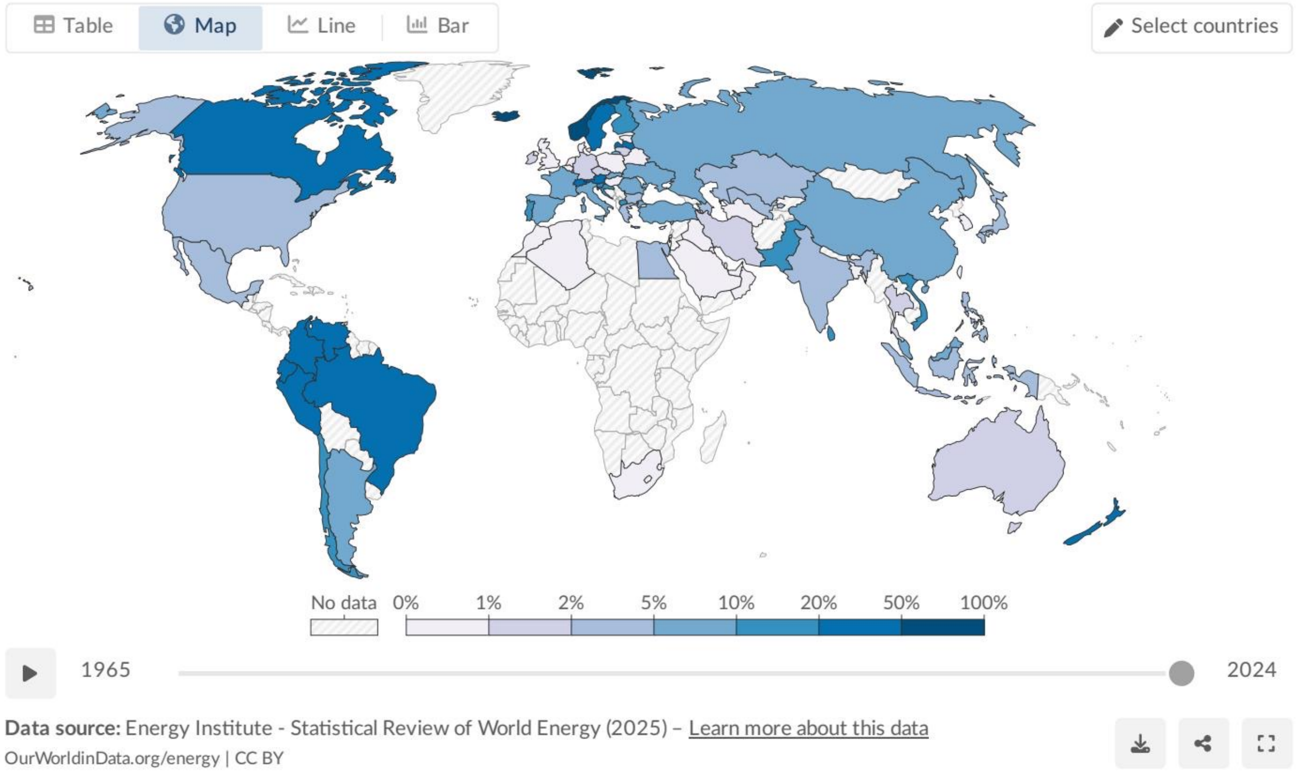
Source: Our World in Data – Renewable Energy | Ember (2026); Energy Institute (2025)

Renewables Now Supply ~14% of Global Primary Energy

~14%

of primary energy

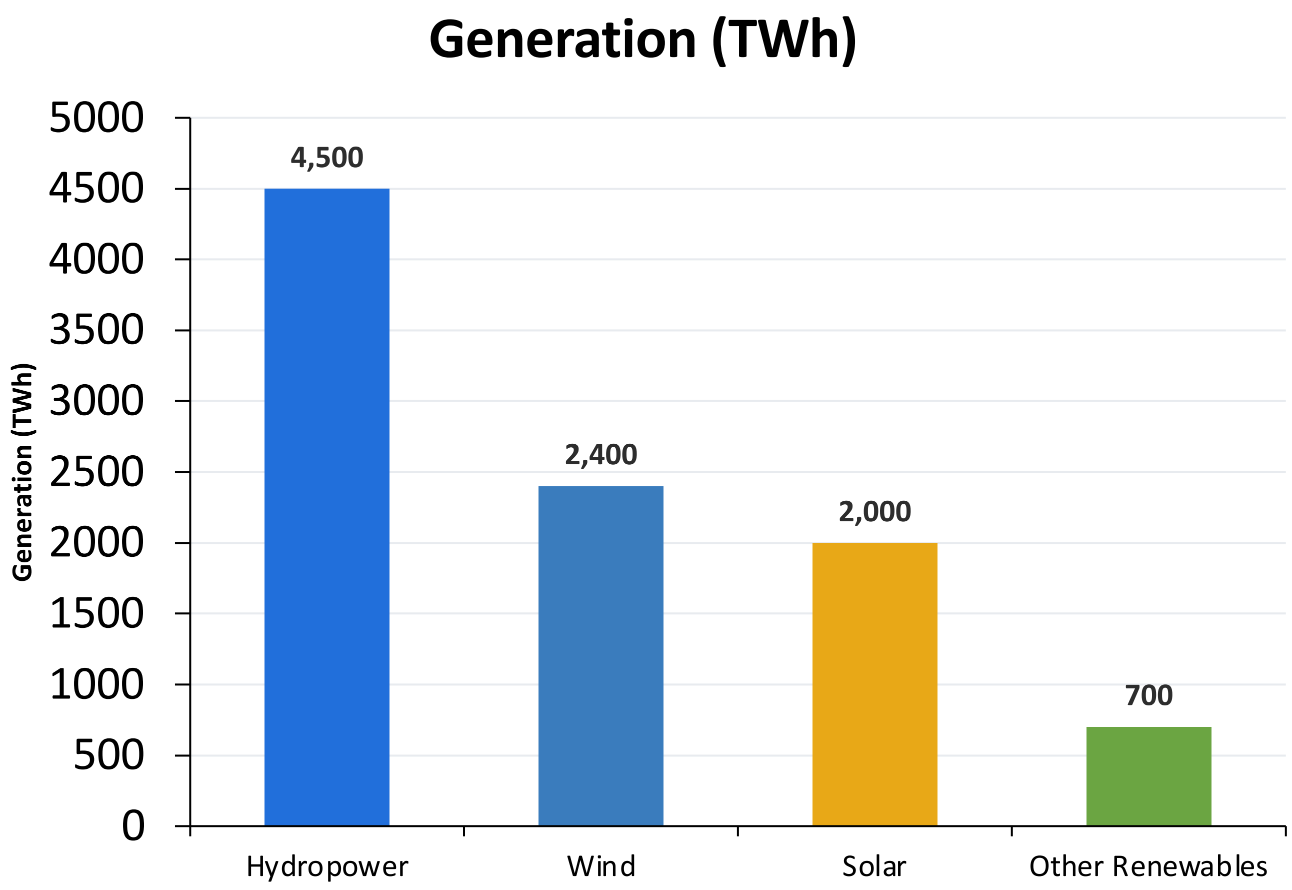
- Approximately one-seventh of the world's primary energy comes from renewable technologies (2024, substitution method).
- Primary energy encompasses electricity, transport, and heating combined.
- Transport and heating are harder to decarbonize, keeping the overall renewable share lower than in electricity alone.
- Norway, Brazil, and New Zealand derive large shares from renewables; Middle East nations remain below 5%.



Share of primary energy from renewables by country, 2024

Source: Our World in Data – Renewable Energy | Ember (2026); Energy Institute (2025)

Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar



Technology	TWh	Share
Hydropower	~4,500	~47%
Wind	~2,400	~25%
Solar	~2,000	~21%
Other	~700	~7%

Wind + Solar Combined

~4,400 TWh (~46% of renewable electricity)

Nearly matching hydropower's output

and growing much faster

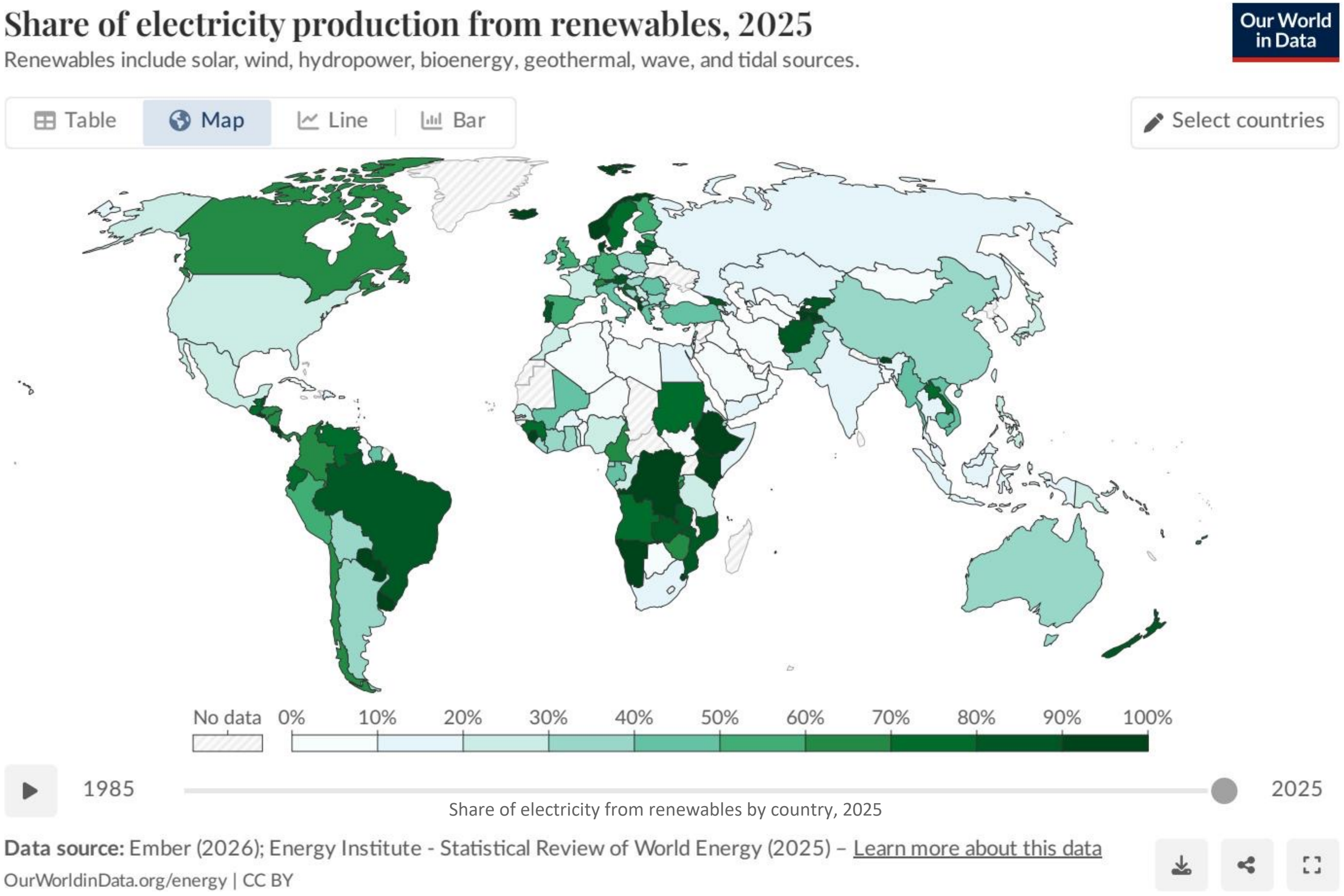
Source: Our World in Data – Renewable Energy | Ember (2026); Energy Institute (2025)

Almost One-Third of Global Electricity Now Comes from Renewables

~30%

of global electricity from renewables

Country/Region	Renewable Share
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%
South Africa	~10%



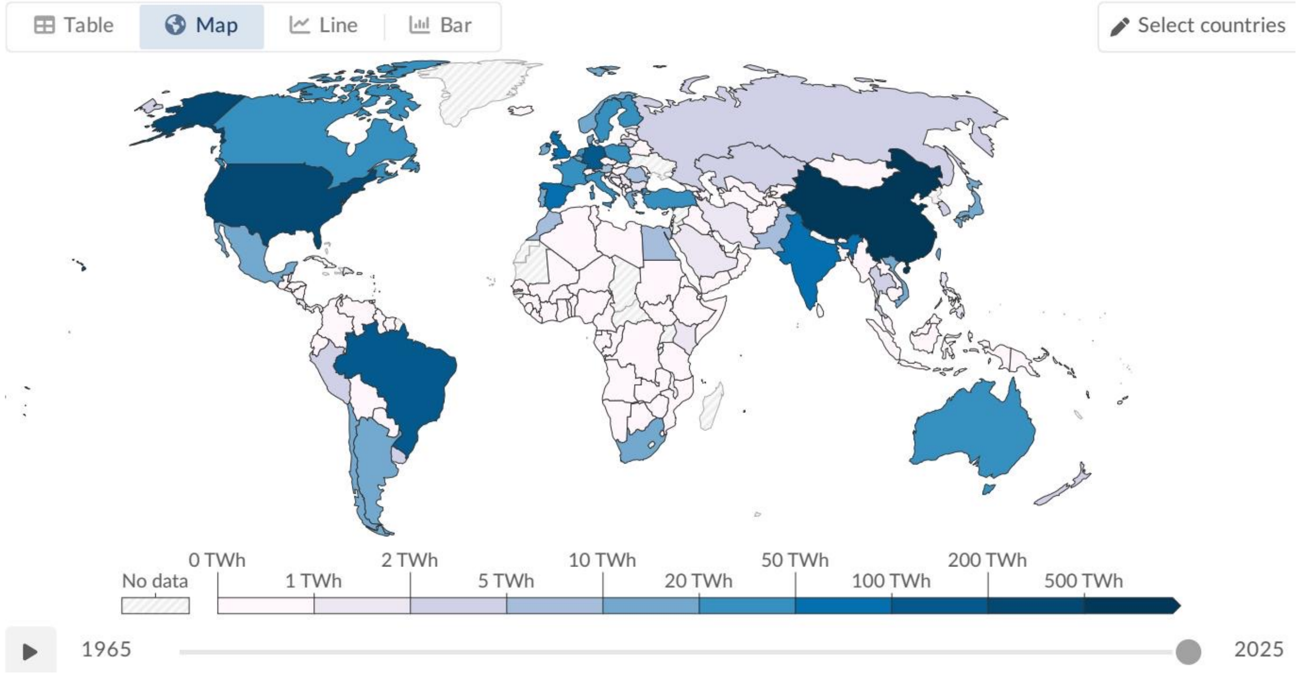
Source: Our World in Data – Renewable Energy | Ember (2026); Energy Institute (2025)

Hydropower Remains the Largest Renewable Source at ~50% of Generation

~50%

of all renewable electricity comes from hydropower

- Hydropower has been generating electricity at scale for over a century.
- Top producers: China, Brazil, and Canada.
- ~4,500 TWh generated globally (2024).
- Hydro's growth potential is limited compared to wind and solar — most suitable sites already developed.
- Despite dominance, hydro's share is declining as wind and solar expand rapidly.



Hydropower generation by country, 2025

Source: Our World in Data – Renewable Energy | Ember (2026); Energy Institute (2025)

Wind and Solar Are the Fastest-Growing Energy Technologies in History

Wind Energy

~2,400 TWh generation (2024)

~25% of renewable electricity

Major contributor in Europe, China, and the US
Includes onshore and offshore installations

Solar Energy

~2,000 TWh generation (2024)

~21% of renewable electricity

Fastest-growing source — dramatic growth since 2010
Driven by falling costs and policy support

Combined: Wind + Solar = ~4,400 TWh — nearly matching hydropower's ~4,500 TWh
and growing much faster than any other energy technology in history

Wind power generation by country, 2025

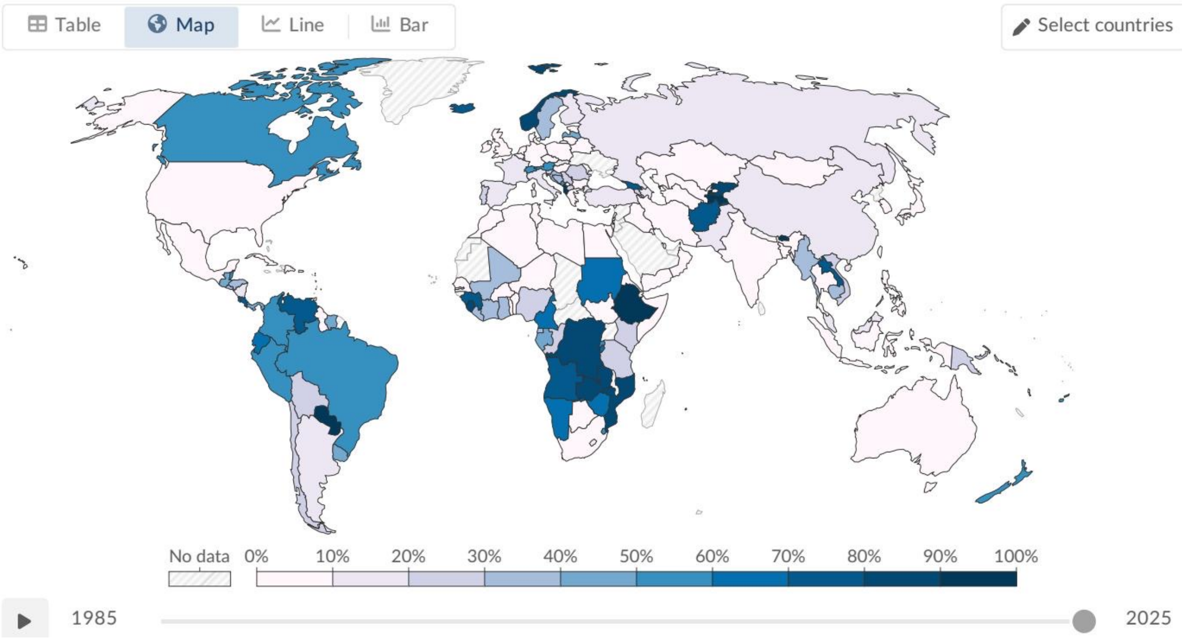
Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and Latin America Lead

Leading Regions

- Scandinavia: 65–98% renewable (abundant hydropower)
- Brazil: ~85% renewable (hydropower)
- Canada: ~65% renewable (hydropower)
- Denmark & Germany: heavy wind & solar investment

Lagging Regions

- Sub-Saharan Africa: minimal renewable infrastructure
- Middle East: almost entirely fossil-fuel dependent
- Saudi Arabia: <2% renewable electricity
- Many African nations: below 10% renewable share



Share of electricity from hydropower by country, 2025

Source: Our World in Data – Renewable Energy | Ember (2026); Energy Institute (2025)

Key Takeaways: Renewables Are Growing Fast but the Transition Must Accelerate

1 **~14% of primary energy, ~30% of electricity**

Renewables have reached a significant milestone, but the overall primary energy share remains low because transport and heating lag behind.

2 **Hydropower still dominant, but wind & solar catching up**

Hydro generates ~50% of renewable electricity, yet wind and solar combined now nearly match hydro's output and are growing far faster.

3 **Wind and solar are the fastest-growing energy technologies in history**

Falling costs and policy support have driven dramatic growth since 2010, with combined output approaching hydropower.

4 **Stark regional disparities persist**

Scandinavia and Latin America exceed 60–90% renewable electricity, while much of the Middle East and Sub-Saharan Africa remain below 10%.

5 **Transport and heating lag far behind electricity**

The renewable share of primary energy is much lower than electricity because transport and heating remain heavily reliant on fossil fuels.

The pace of the renewable transition must accelerate significantly to meet climate targets and close regional gaps.